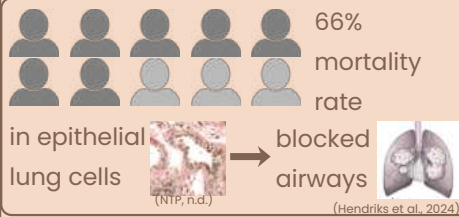


Determining Whether Bilobalide Can Enhance Curcumin's Ability to Induce p53-Dependent Cell Death in Non-small Cell Lung Adenocarcinoma

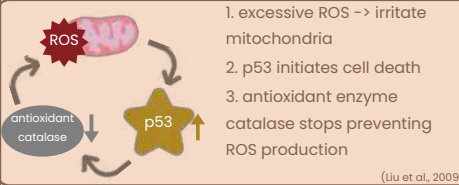
Laila Mogassbi

Introduction

NSCLC



Positive-Feedback Mechanism



Treatments

- synthetic nitrogen mustard-bendamustine: has common vomiting & alopecia side effects
 - newly explored phytochemicals: show promise as affordable and safer alternatives
- (Amjad, M., 2023)

Bilobalide (Liu et al., 2021)

Ginkgo biloba
killed gastric cancer

Curcumin (Ye et al., 2015)

Curcuma longa
p53 in NSCLC

Project Overview

Research Question

Can bilobalide enhance curcumin's p53-dependent cell death ability in NSCLC?

Hypothesis

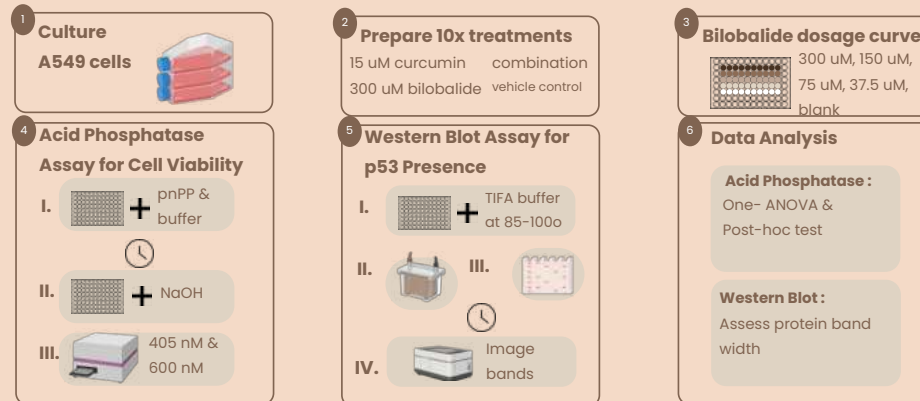
curcumin + bilobalide → p53 ↑ cell viability ↓

Justification

bilobalide induced excessive ROS → excessive ROS induced p53 dependent apoptosis

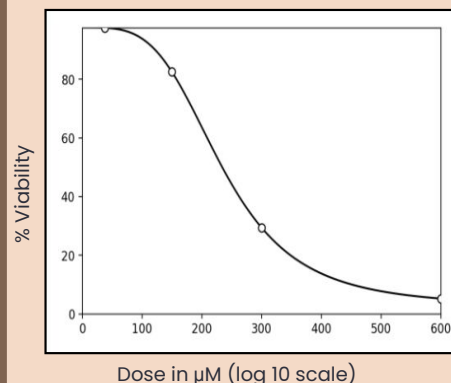
(Liu et al., 2009)

Methodology



Results

Bilobalide Dosage Curve



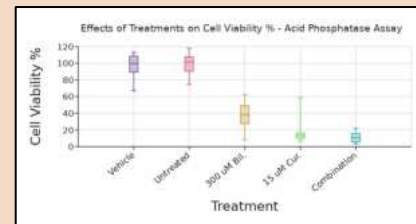
Cell Viability Assay

3 trials, n=27 to n=30 wells for all groups

ANOVA Test P-value: <0.01

Post-Hoc Test

Pair	Cur. - Bil.	Com. - Bil.	Veh. - Bil.	Untr. - Bil.	Com. - Cur.
P-value	<0.01	<0.01	<0.01	<0.01	0.65
Pair	Veh. - Cur.	Unt. - Cur.	Veh. - Com.	Untr. - Com.	Untr. - Veh.
P-value	<0.01	<0.01	<0.01	<0.01	0.99



Expected Western Blot

Highest p53 in combination Medium in individual curcumin/bilobalide Lowest in vehicle/- controls

Discussion

Dosage Curve

Bilobalide IC50 value of 234.1 μM

Close to 300 μM value found by past studies (Liu et al., 2021)

Cell viability

Although the p-value between groups is <0.01, making them significantly different, the product of the mean viabilities of curcumin and bilobalide treated cells is $36.587\% \times 14.689\% = 5.374\%$, which is more than the combination treatment cells' mean viability of 10.152%.

Combination is **antagonistic**, meaning that it **decreases the effects of bilobalide and curcumin**.

Hypothesis is **not supported**

Implications

Bilobalide may serve significant therapeutic effects on NSCLC by decreasing cell viability through unknown mechanisms that do not align with curcumin

Future Work

- Determining cell death type - apoptosis or not
- Testing ROS of NSCLC when treated with bilobalide directly
- Exploring other bilobalide mechanisms

References

- Hendriks, L.E.L., Remon, J., Faivre-Finn, C., Guassino, M., Heymach, J., Kerr, K., Tan, D., Veronesi, G., Rack, M. (2024). Non-small-cell lung cancer. *Nature Reviews Disease Primers*, 10(71), 1-22. <https://doi.org/10.1038/s41572-024-20551-9>
- Liu, B., Chen, Y., & St Clair, D. K. (2008). ROS and p53: a versatile partnership. *Free radical biology & medicine*, 44(8), 1529-1535. <https://doi.org/10.1016/j.freeradbiomed.2008.01.011>
- Liu, J., Geng, Z., Zhang, Y., Alharbi, S., Shi, Y. (2021) Sesquiterpenoid bilobalide inhibits gastric carcinoma cell growth and induces apoptosis both in vitro and in vivo models. *Journal of Biochemical and Molecular Toxicology*, 35(5), 22723. doi.org/10.1002/jbt.22723
- Nonneoplastic Lesion Atlas. (n.d.) *Lung, Epithelium - Degeneration*. National Toxicology Program. https://ntp.niehs.nih.gov/sites/ntp/registries/ntplib/Lung_Epithelium-Degeneration
- Ye, M., Zhang, J., Zhang, J., Miao, Q., Yao, L., Zhang, J. (2015). Curcumin promotes apoptosis by activating the p53-miR-192-Sp215-XIAP pathway in non-small cell lung cancer. *Cancer Letters*, 357(1), 196-205. doi.org/10.1016/j.canlet.2014.11.028